SUBSTATION No. 68
CARBON IN LEACH (C.I.L. 3)

SUBSTATION EARTH BAR

415V FROM CIL 3 MCC
30-SW-62
2000-F30-SLD-EL-10101

F30-SW-401
CIL 3 415V MOTOR CONTROL CENTRE

MCC EARTH BAR

NEUTRAL BAR
NOTE 1

415V 3NE~ 50Hz 800A 63kA for 1sec

1F-05	1F-08	1F-11	1F-14	1F-17	1F-21	2F-17	2F-21	4F-19	3F-05
100A	100A	100A	100A	250A	250A	160A	160A	160A	250A
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >	/ >
100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	100/1A	90A	100/1A
								EOL	
M 3~	M 3~	M 3~	M 3~	t° M 3~	t° M 3~	t° M 3~	t° M 3~	t° M 3~	t° M 3~
30kW	30kW	30kW	30kW	75kW	75kW	55kW	55kW	55kW	75kW
F30-PP-415 CONCENTRATE PRE-LEACH THICKENER U/F PUMP 1 2000-F30-SCM-EL-10001	F30-PP-416 CONCENTRATE PRE-LEACH THICKENER U/F PUMP 2 2000-F30-SCM-EL-10002	F30-PP-417 CONCENTRATE THICKENER UNDERFLOW PUMP 1 2000-F30-SCM-EL-10017	F30-PP-418 CONCENTRATE THICKENER UNDERFLOW PUMP 2 2000-F30-SCM-EL-10018	F30-PP-426 CONCENTRATE LEACH TRANSFER PUMP 1 2000-F30-SCM-EL-10003	F30-PP-427 CONCENTRATE LEACH TRANSFER PUMP 2 2000-F30-SCM-EL-10004	F30-PP-428 CIL 3 TAILS PUMP 1 2000-F30-SCM-EL-10005	F30-PP-429 CIL 3 TAILS PUMP 2 2000-F30-SCM-EL-10006	F30-PP-476 CIL 3 TANK A LOADED CARBON RECOVERY PUMP 2000-F30-SCM-EL-10029	F30-PP-496 ACID EFFLUENT AND FINE CARBON PUMP 1 2000-F30-SCM-EL-10020

NOTE 7

F30-PP-576
ALTERNATIVE LOCATION
CIL 3 TANK B LOADED
CARBON RECOVERY PUMP
2000-F30-SCM-EL-10029

CONTINUED ON SHT 2
2000-F30-SLD-EL-10302

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM 30-SW-62 INCOMER. SIGNAGE SHALL BE PLACED ON F30-SW-401 INCOMING SUPPLY SECTION READING "M.E.N. LINK LOCATED IN 30-SW-62".
2. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
3. ALL DRIVE I/O VIA COMMS NETWORK TO DCS.
4. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
5. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
6. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
7. SECOND DECONTACTOR INSTALLED ABOVE CIL3 TANK B, AT THE F30-PP-476 ALTERNATE LOCATION. FLEX CABLE IS INSTALLED BETWEEN THE CIL3 TANK A AND CIL3 TANK B DECONTACTORS, WITH A PLUG LOCATED AT THE CIL3 TANK A DECONTACTOR END. CONNECT THE PLUG TO THE CIL3 TANK A DECONTACTOR ONLY WHEN F30-PP-476 IS LOCATED AT CIL3 TANK B.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 2, 3
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 2, 3
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 2, 3
- THERMAL OVERLOAD UNIT NOTE 4
- VARIABLE SPEED DRIVE NOTE 2, 3
- / > — EARTH LEAKAGE RELAY NOTE 2, 3
- FIELD-MOUNTED DECONTACTOR
- t° — MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 6)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

NORTHERN STAR

Approved	☒
Approved as Noted	☐
Not Approved	☐
For Information Only	☐

Signed [redacted] Date 05/12/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F30-SLD-EL-10301_Rev1

Carbon in leach (CIL3) substation no. 68 MCC
F30-SW-401 single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F30 - CIL2_3 & CONCENTRATE THICKENING\2000-F30-SLD-EL-10301.DWG 19NOV25 11:45AM HOTSEAT04